

Harrison F. Squires

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RESEARCH EXPERIENCE

Experimental Biophysics PhD

September 2026 – Present

University of Edinburgh

Edinburgh, UK

- Project: “Towards machine olfaction, a new ISR modality”, supervised by Prof. Davide Michieletto (academic) and Dr Neal Hopkins (industry; Dstl).
- Research focus: nucleic acid nanotechnology.
- Recipient of SPADS CDT doctoral funding.
- Awarded an additional £10k annual Research Training Support Grant from the OCSA Bio Tech Centre.

Bridging Research Intern

June 2026 – August 2026

University of Edinburgh

Edinburgh, UK

- Awarded bridging funding to undertake research before commencing PhD studies.
- Project: “DNA Nanostars”, supervised by Prof. Davide Michieletto and Dr Jenny Harnett.
- Investigated liquid–liquid phase separation in DNA nanostars and its potential application in lanthanide capture.

MPhys Research Project

September 2025 – May 2026

University of Edinburgh

Edinburgh, UK

- Project: “Evaporation of Breath Figures”, supervised by Dr David Fairhurst and Dr Joey Kilbride.
- Extended a recently published study by investigating the dynamics of the macroscopic edge effect in respiratory breath figures and its relationship to droplet-size distribution.
- Designed and built an experimental setup for imaging large droplet arrays.
- Developed efficient image, data, and statistical analysis workflows using FIJI and Python.

Summer Research Intern

June 2025 – August 2025

QinetiQ / University of Edinburgh

Edinburgh, UK

- Produced “Terahertz Methods for the Characterisation of Materials”, a literature review advising the company’s future research direction.
- Summarised findings in a final report and developed a comparative table of practical trade-offs.
- Funded by the University of Edinburgh Career Development Scholarship and hosted by QinetiQ.

Research Intern

July 2024 – June 2025

University of Melbourne / Carbon Cybernetics

Melbourne, Australia

- Project: “On the Characterisation of Carbon Fibre Electrodes for Neural Recording”, supervised by Prof. Steven Prawer.
- Contributing author on *Modelling of Intracortical Microwire Electrodes for Brain–Machine Interfaces* (manuscript in preparation).
- Designed and built a benchtop testing system to optimise carbon-fibre microelectrode-array performance.
- Undertook a meta-analysis of *in vivo* datasets to inform optimal array design.
- Automated data acquisition and analysis workflows using MATLAB and Python.
- Delivered weekly research presentations, incorporating group feedback into further development.

EDUCATION

University of Edinburgh

2021 – 2026

Master of Physics (MPhys), Physics with a Year Abroad — First-Class Honours (75%)

Edinburgh, UK

- Received the Margaret Campbell-Scott Scholarship for outstanding entry qualifications.
- Awarded the Career Development Scholarship for a summer project in conjunction with QinetiQ.
- Achieved Certificates of Merit for academic excellence in the first two years.
- Awarded the Junior Honours Class Medal for top performance in the third-year cohort.
- Earned the Edinburgh Award for volunteering contributions to the University Sports Union Jiu Jitsu Club.
- Final-year focus: soft condensed matter (experiment and theory), biological physics, and statistical physics.

EMPLOYMENT HISTORY

Academic Tutor

March 2025 – June 2025

Newman College, University of Melbourne

Melbourne, Australia

- Tutored first- and second-year Bachelor of Science students in *Foundations of Physics (PHYC10009)* and *Quantum and Thermal Physics (PHYC20012)*.
- Ran individual consultations and group tutorials, tailoring support to students' areas of difficulty.
- Designed original resources and problem sets to present physics concepts in novel contexts.

Cheesemonger

June 2023 – December 2023

IJ Mellis

Edinburgh, UK

- Independently managed shop operations, including opening and closing responsibilities.
- Completed specialised training in product knowledge and food safety.
- Developed expert knowledge of artisanal and regional cheeses.

VOLUNTEERING

English as a Second Language Tutor

March 2025 – June 2025

Jesuit Worldwide Learning

Remote

- Supported a refugee student in Kenya in developing English-language skills through personalised, one-to-one conversation practice.